

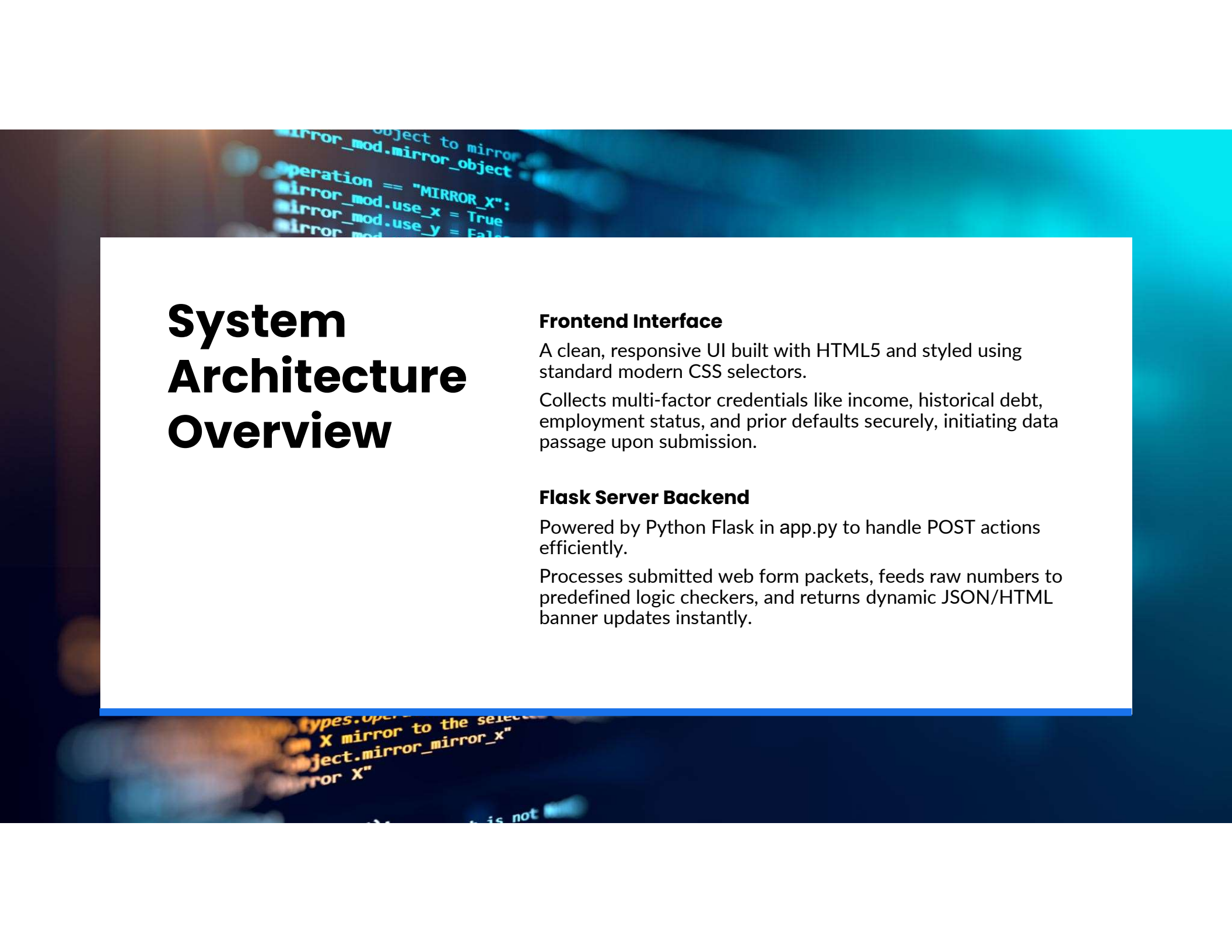
Credit Card Approval Prediction System

An automated web portal processing user credentials in real-time to compute instant risk evaluations and card approvals.

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TEAM ID: SWTID-2026-5556





System Architecture Overview

Frontend Interface

A clean, responsive UI built with HTML5 and styled using standard modern CSS selectors.

Collects multi-factor credentials like income, historical debt, employment status, and prior defaults securely, initiating data passage upon submission.

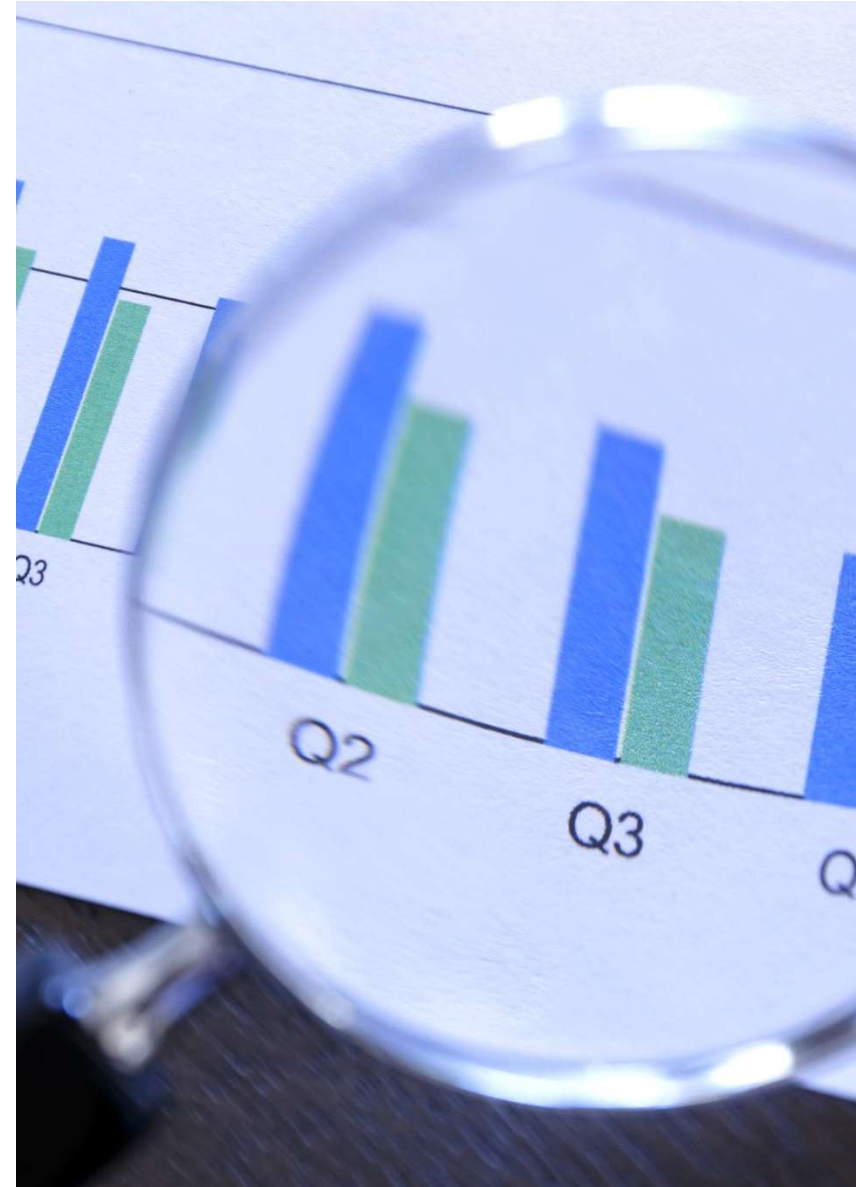
Flask Server Backend

Powered by Python Flask in app.py to handle POST actions efficiently.

Processes submitted web form packets, feeds raw numbers to predefined logic checkers, and returns dynamic JSON/HTML banner updates instantly.

Decision Evaluation Criteria

- **Income & Debt**
 - Verifies the client's current annual income bracket versus existing debt levels to gauge structural repayment capacity.
- **Employment**
 - Calculates the number of months the customer has spent employed. Continuous work history reduces risk profiles.
- **Credit History**
 - Scans historical records for negative indicators like prior bankruptcies, payment defaults, or ongoing legal alerts.





Performance Metrics

- **<50ms**
- **API LATENCY TIME**
- **High-Performance Local Framework**
- By using structured conditional routing in Python instead of heavy sequential server databases, we achieved stellar load performance:
- **Zero Database overhead:** Ideal for rapid proof-of-concept testing.
- **Instantly Responsive UI:** Dynamic visual cards load within milliseconds on browsers.
- **Optimized Memory:** Lightweight server profile executes easily on any local BTech hardware.



Future Scope: Machine Learning

Transitioning to ML Models

Moving beyond hardcoded thresholds to deep, data-driven analytical model structures:

- **Dataset Training:** We plan to use our `creditcard.csv` file containing historical client records.
- **Predictive Accuracy:** Implementing classification algorithms such as Logistic Regression or Random Forests.
- **Dynamic Scale:** Storing user responses inside standard databases to retrain our predictor over time.



Thank You